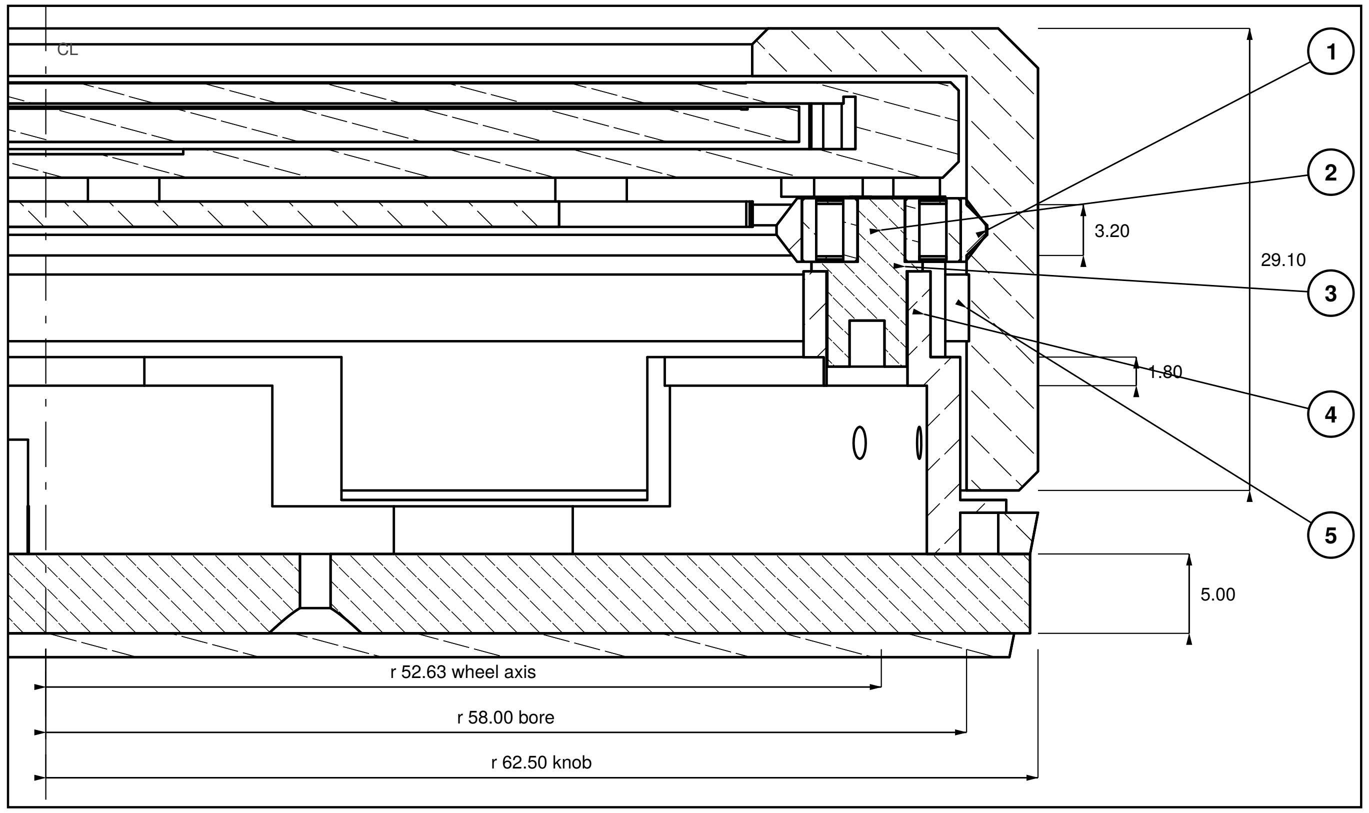
SECTION A–A scale 2.1 : 1



Right half, on the 30°–210° axis, through the wheel at 30°. Hatching is cut material; plain lines are behind the cut.

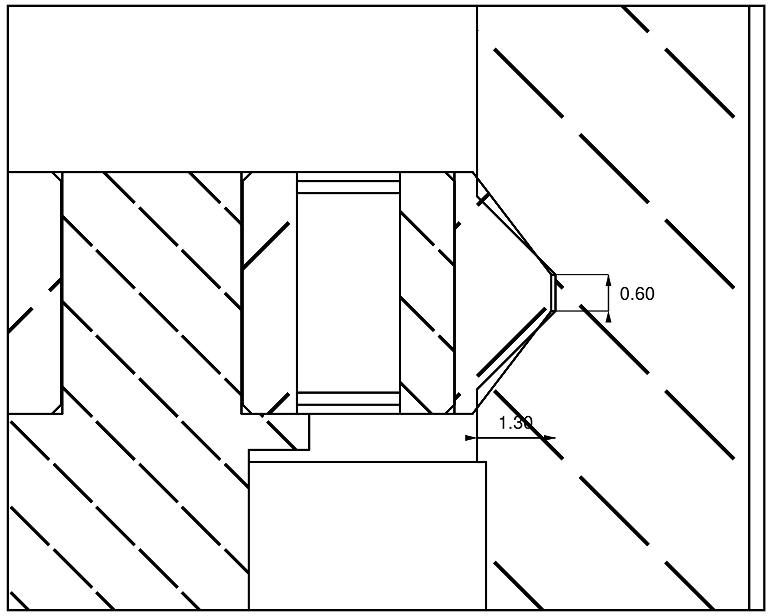
NOTES

1. V-groove in the bore: root r 59.30, 90° included, 1.30 deep, 0.60 flat at the root so contact is on the flanks only. z 23.8–27.0. The bore is Ø116 and smooth elsewhere, apart from the 0.15 code-band recess at z 18.4–22.6 (D02).
2. 623ZZ bearing 3 × 10 × 4 pressed into a turned Ø13.20 V-collar. Axis r 52.63, z 23.4–27.4. Three off, at 30° / 150° / 270°.
3. Eccentric bush: a Ø5.00 shank in the post carrying a Ø2.96 wheel pin offset 0.90 from it, under a Ø7.00 flange 0.60 thick, with a 2 mm hex socket underneath. Pin inward, the wheel sits at r 50.83 — 0.57 clear inside the bore — and the knob drops on; half a turn from below seats it and sets the preload.
4. Wheel post Ø8.00 on the ledge (z 15.6–17.4), Ø5.10 bush bore, top face z 22.8. v9 moved the M3 plate screws to 125° / 245° / 332°, so post and screw no longer share structure: the posts stand on the ledge alone.
5. The wheels carry the knob's weight and its lift-off load. Nothing else supports it: the drive band is a friction contact on the same bore, and the halo, the code band and the display all clear the knob.

V-on-V locates the knob radially and axially at once and takes the lift-off load, so the knob cannot be pulled off the device.

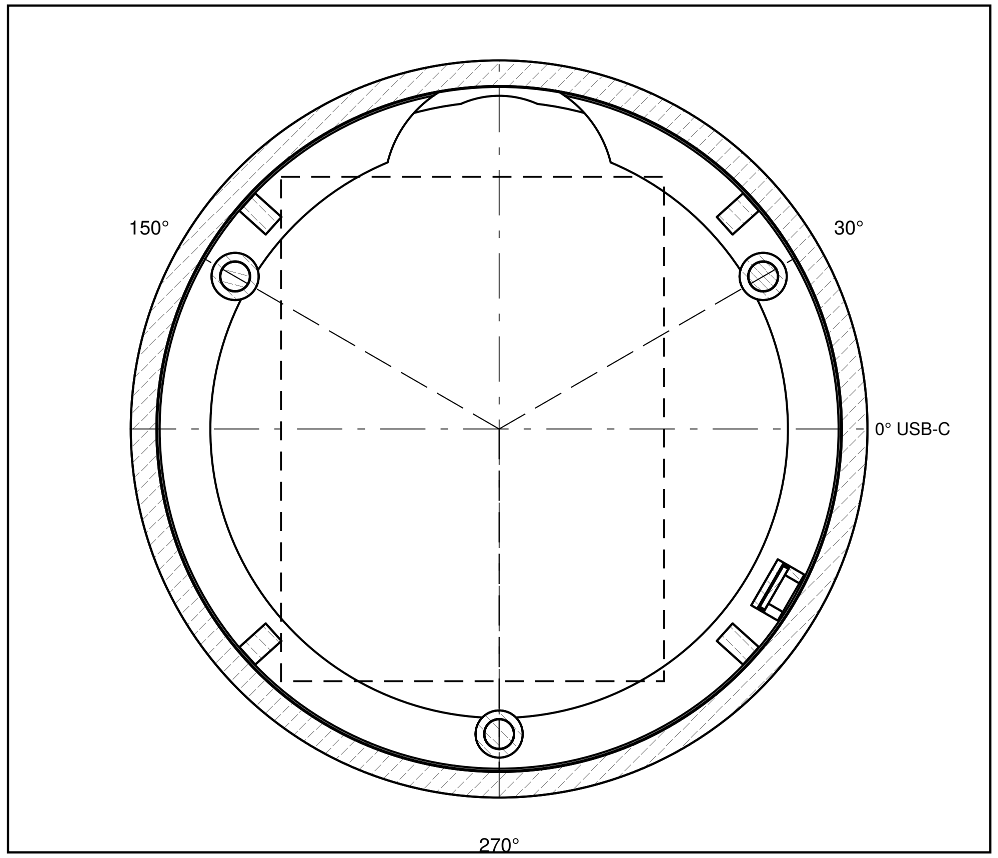
Knob 29.1 tall: 76 % of the 38.1 body height, 73 % of the 39.6 including the pad.

DETAIL X — GROOVE AND COLLAR 8 : 1



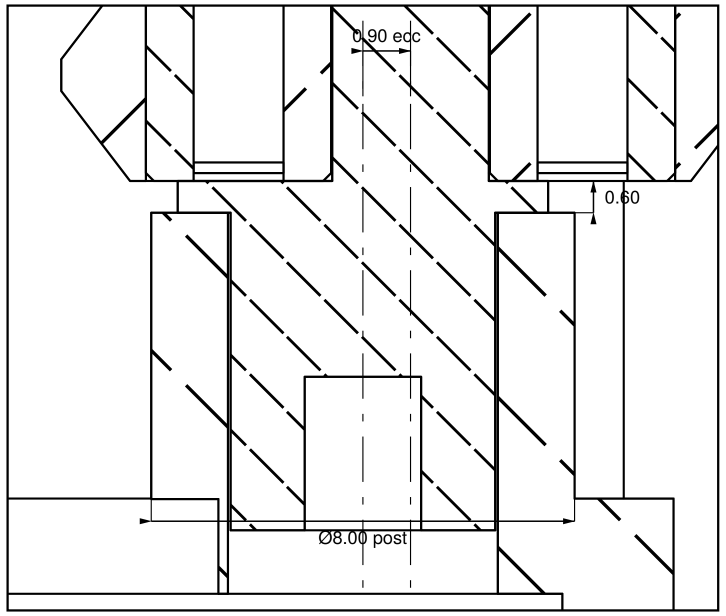
Notchiness three times a turn means the collars are bottoming on the 0.60 root flat. Bore-to-groove concentricity within 0.05 total: runout modulates the wheel preload, and the wheels are what hold the knob.

PLAN — HORIZONTAL SECTION AT z 20 scale 0.78 : 1



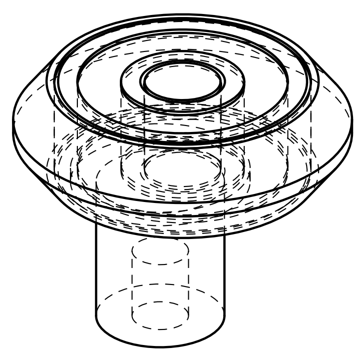
Dashed rectangle: the display board, 65.0 × 85.5, centre 4.5 in −x — its real outline from the model, above this cut.
The board's edge reaches r 42.76 at 270°, the worst of the three; the post's inner face is at r 47.73, so 4.97 clear. That is why the wheels are at 30° / 150° / 270°.
The lone block at 330° is the encoder tower; the four at 42° / 138° / 222° / 318° are the display columns. Both are on D02.

DETAIL Y — BUSH AND POST 7 : 1



Bush flange 0.60 thick sits on the post top at z 22.8. Printed pin is the known weak point — brass with an M2 grub screw in production.

WHEEL AND BUSH — ISOMETRIC 3.4 : 1



V-collar over the 623ZZ, on the eccentric bush, as assembled. Hidden edges dashed.

MODELLED, NOT PRINTED — no v9 part has been made. Dimensions are model values, not inspection results. Bodies tagged ENVELOPE or ASSUMED in the model are drawn as envelopes, not as the real part.

Projected from docs/v9/the60_v9/src/params.py + model.py (build123d) by hidden-line projection. No hand drafting; dimension text is written from the parameter values.

KNOB SUPPORT — WHEELS, COLLARS AND ECCENTRIC BUSHES

the 60 (Cadrane) — desk dial	DRAWING 60-09-D01	REV v9
SCALE AS SHOWN	DATE 2026-09-04	UNITS mm